

Generative Fashion Designer

Advanced Deep Neural Networks for Textile Pattern Generation

Team

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This report focuses on system design, datasets, models, results, metrics, and deployment.
Implementation details and source code are intentionally omitted.

Executive Summary

This document presents the design and evaluation of the Generative Fashion Designer system. It emphasizes engineering decisions, evaluation metrics, and deployment considerations. The report avoids including source code; the repository contains all implementation artifacts.

Overview & Objectives

- Deliver a modular generative system for textile pattern synthesis.
- Provide reproducible pipelines and quantitative evaluation.
- Package models for serving via a REST API and demo UI.

Architecture

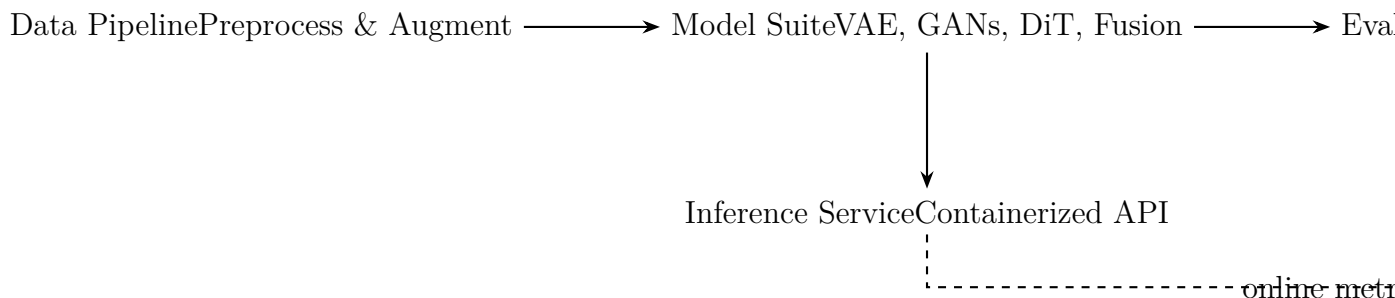


Figure 1: System architecture: data $\rightarrow$ models $\rightarrow$ evaluation $\rightarrow$ inference

Data

The primary dataset and preprocessing steps are documented: standardized image sizes, normalization, and augmentations to improve robustness and diversity. Artifacts are versioned with hashes for reproducibility.

Models & Training

Multiple generative families were developed and tuned. Training best practices include mixed precision when available, scheduler tuning, and model selection based on validation FID and visual inspection.

Evaluation and Metrics

We evaluate using FID, Inception Score (IS), and SSIM. Representative results are summarized below.

Table 1: Representative results on 64×64 test set

	Model	FID (↓)	IS (↑)	Remarks
textwidth	VAE	42.5	3.2	Interpretable latent
	DCGAN	38.2	4.1	Sharp output
	WGAN-GP	35.7	4.5	Stable training
	Latent DiT	32.1	4.8	Best distribution

Deployment and Infrastructure

Models are checkpointed with metadata and served via a containerized REST endpoint; the deployment includes simple monitoring and gallery generation for qualitative review.

Design Decisions & Lessons

Key engineering choices: dataset versioning, modular trainers per architecture, and conservative augmentation to preserve cultural characteristics.

Appendix: Artifacts

Generated galleries, checkpoints, and the inference server configuration are present in the repository under the ‘outputs’ and ‘checkpoints’ directories.

References

- [1] Goodfellow, I. et al. Generative Adversarial Nets. 2014.
- [2] Ho, J., Jain, A., & Abbeel, P. Denoising Diffusion Probabilistic Models. 2020.